

Medvale Internal Medicine

Chapter 3G: Biliary Tract, Gallbladder, Pancreatic Disease, and Appendix

Original study text for clinical reasoning, numeric anchors, and graph-RAG extraction

Scope. This chapter organizes disorders that present with right upper quadrant pain, epigastric pain, jaundice, abnormal cholestatic liver tests, sepsis from biliary obstruction, acute pancreatitis, chronic pancreatic insufficiency, pancreatic malignancy, and appendicitis. The emphasis is not memorizing isolated disease names; it is linking anatomy, duct obstruction, inflammation, infection, ischemia, and operative urgency into a reproducible diagnostic map.

Coverage note: The source packet for this segment contained gallbladder and biliary tract disease, primary sclerosing cholangitis, primary biliary cholangitis, cholangiocarcinoma, choledochal cysts and strictures, biliary dyskinesia, and appendicitis. Additional pancreas coverage is included to complete the planned hepatobiliary-pancreatic chapter.

1. Why This Matters Clinically

Hepatobiliary and pancreatic disease is high-yield because several syndromes look similar at first contact but have very different time courses. A patient with post-prandial right upper quadrant pain and a normal white count may need elective outpatient cholecystectomy; a patient with fever, jaundice, hypotension, and a dilated common bile duct needs resuscitation, antibiotics, and urgent biliary decompression. A patient with epigastric pain and a lipase elevation may need supportive pancreatitis care, whereas the same pain plus obstructive jaundice in an older adult raises concern for pancreaticobiliary malignancy.

For internal medicine decision-making, the recurring question is whether the process is mechanical obstruction without infection, inflammatory obstruction, infected obstruction, pancreatic autodigestion, or surgical abdomen. The history, vital signs, bilirubin pattern, alkaline phosphatase, transaminases, white count, lactate, lipase, and imaging sequence usually clarify which branch the patient occupies.

2. Core Anatomy and Pattern Recognition

Bile is made by hepatocytes, modified by cholangiocytes, stored in the gallbladder, and released into the duodenum after cholecystokinin stimulates gallbladder contraction and sphincter of Oddi relaxation. Stones can remain silent in the gallbladder, intermittently obstruct the cystic duct, lodge in the common bile duct, or block the ampulla where the pancreatic duct joins the biliary tree. This explains why a single gallstone pathway can produce biliary colic, cholecystitis, choledocholithiasis, cholangitis, or gallstone pancreatitis.

Original diagram: gallstone complication pathway

Gallstones	Cystic duct block	Cholecystitis
Gallstones	Common bile duct stone	Jaundice / cholangitis
Gallstones	Ampullary obstruction	Gallstone pancreatitis

A cholestatic lab pattern means bile flow is impaired. In extrahepatic obstruction, alkaline phosphatase and gamma-glutamyltransferase often rise disproportionately compared with AST and ALT, conjugated bilirubin can produce dark urine, and reduced bile pigment delivery to the gut can cause pale or clay-colored stool. Early obstruction can temporarily cause high aminotransferases, but persistent duct obstruction usually declares itself as a cholestatic pattern.

Original diagram: jaundice and liver test pattern map

Pattern	Typical labs	Examples
Hepatocellular	AST/ALT high relative to alkaline phosphatase	Viral hepatitis, ischemic hepatitis, toxins, alcohol
Cholestatic	Alkaline phosphatase/GGT and direct bilirubin high	CBD stone, stricture, pancreatic cancer, PSC, PBC
Mixed	Both transaminases and cholestatic markers elevated	Drug injury, sepsis, advanced obstruction

Clinical pattern	Most likely process	Helpful first test	Immediate priority
Episodic RUQ/epigastric pain after meals, no fever, normal labs	Biliary colic from transient cystic duct obstruction	RUQ ultrasound	Analgesia; outpatient surgery referral if recurrent
Constant RUQ pain >4-6 h, fever/leukocytosis, Murphy sign	Acute cholecystitis	RUQ ultrasound; HIDA if equivocal	Admit, NPO, IV fluids, antibiotics, early cholecystectomy
Jaundice + cholestatic labs, maybe minimal pain	Choledocholithiasis or malignant obstruction	RUQ ultrasound then MRCP/EUS/ERCP by risk	Define obstruction; ERCP if high-risk stone or cholangitis
Fever + jaundice + RUQ pain; hypotension/AMS if severe	Ascending cholangitis	Labs/cultures, ultrasound, ERCP/PTC	Sepsis care, antibiotics, urgent decompression
Epigastric pain radiating to back + lipase $\geq 3 \times$ ULN	Acute pancreatitis	Lipase; ultrasound for gallstones	Analgesia, monitored fluids, early feeding when tolerated
Migratory periumbilical to RLQ pain, anorexia, localized tenderness	Appendicitis	CT in adults if uncertain; ultrasound in selected patients	Surgery consult; antibiotics; appendectomy or selected nonoperative pathway

3. Cholelithiasis and Biliary Colic

Illness script

Cholelithiasis means gallstones are present in the gallbladder. Many patients remain asymptomatic for years. When symptoms occur, the classic syndrome is biliary colic: episodic right upper quadrant or epigastric pain caused by transient cystic duct obstruction during gallbladder contraction. Pain is often post-prandial, especially after fatty meals, and may radiate to the right scapula or shoulder. Nausea is common. Because obstruction is transient and the gallbladder wall is not yet persistently inflamed, fever, sustained leukocytosis, and peritoneal signs should be absent.

Cholesterol stones are associated with obesity, diabetes, pregnancy, estrogen exposure, rapid weight loss, age, and some ethnic backgrounds. Pigment stones reflect bilirubin load or biliary infection; black pigment stones are linked to chronic hemolysis and cirrhosis, while brown pigment stones more often form in infected ducts. Mixed stones are common in practice and explain why stone composition is usually less important clinically than the anatomic location of obstruction.

Stone/process	Mechanism	Risk associations	Clinical implication
Cholesterol stone	Supersaturation of bile with cholesterol plus gallbladder stasis	Female sex, pregnancy, obesity, diabetes, rapid weight loss, estrogen exposure	Most common stone type; often causes biliary colic or cholecystitis
Black pigment stone	Calcium bilirubinate precipitation from unconjugated bilirubin load	Hemolysis, cirrhosis	Think chronic hemolysis or advanced liver disease
Brown pigment stone	Biliary infection/stasis with bacterial deconjugation	Biliary strictures, parasites, recurrent duct infection	More likely in ducts; linked to cholangitis/choledocholithiasis
Biliary sludge	Microlithiasis and crystals without large stones	Pregnancy, critical illness, TPN, rapid weight loss	Can cause pancreatitis or biliary pain despite subtle imaging

Diagnosis and management

Right upper quadrant ultrasound is the usual first test because it identifies gallstones, gallbladder wall thickening, pericholecystic fluid, sonographic Murphy sign, and biliary duct dilation. For uncomplicated biliary colic, laboratory tests may be normal. Recurrent biliary colic is usually managed with elective laparoscopic cholecystectomy, because symptoms often recur and future complications can include acute cholecystitis, choledocholithiasis, cholangitis, pancreatitis, gallstone ileus, or gallbladder malignancy. Asymptomatic stones usually do not require treatment, except in select high-risk situations such as porcelain gallbladder or very high-risk surgical planning contexts.

4. Acute Cholecystitis

Pathophysiology and presentation

Acute calculous cholecystitis begins with persistent cystic duct obstruction. Chemical inflammation and gallbladder distention produce constant pain, and bacterial infection may develop secondarily. The pain of biliary colic usually lasts minutes to a few hours; the pain of acute cholecystitis typically persists beyond 4 to 6 hours and is accompanied by fever, leukocytosis, anorexia, nausea, vomiting, and right upper quadrant tenderness. A positive Murphy sign occurs when inspiratory arrest is triggered by palpation over the inflamed gallbladder.

Acalculous cholecystitis occurs without an obstructing stone and is classically seen in critically ill, postoperative, burn, trauma, or septic patients. Because these patients may be sedated or unable to localize pain, the diagnosis is frequently delayed. Unexplained fever, leukocytosis, sepsis, or cholestatic labs in a critically ill patient should trigger consideration of acalculous disease.

Feature	Biliary colic	Acute cholecystitis	Acalculous cholecystitis
Primary problem	Transient cystic duct obstruction	Persistent cystic duct obstruction with inflammation	Gallbladder ischemia/stasis in critical illness
Pain duration	Usually <4-6 h	Persistent, often >4-6 h	May be occult or nonspecific
Systemic signs	Absent	Fever, leukocytosis common	Fever/sepsis may dominate
Ultrasound	Stones; no marked wall inflammation	Stones, wall thickening, pericholecystic fluid, Murphy sign	Wall thickening/sludge; stones absent
Treatment	Elective cholecystectomy if recurrent	Admit, NPO, IV fluids, antibiotics, early cholecystectomy	Antibiotics + urgent drainage/cholecystostomy or surgery

Diagnostic approach

Ultrasound is first-line. A HIDA scan is useful when ultrasound is equivocal and suspicion remains high; nonvisualization of the gallbladder supports cystic duct obstruction. CT is not the preferred first test for uncomplicated cholecystitis but is useful when complications or alternative diagnoses are possible, such as perforation, abscess, pancreatitis, malignancy, or bowel pathology.

Complications include gangrenous cholecystitis, emphysematous cholecystitis, perforation, pericholecystic abscess, empyema of the gallbladder, cholecystoenteric fistula, and gallstone ileus. Diabetes and vascular disease raise concern for emphysematous or gangrenous disease. Free air, severe toxicity, peritonitis, or persistent sepsis should prompt urgent surgical evaluation.

Management

Initial management includes admission, NPO status, IV fluids, analgesia, correction of electrolyte abnormalities, and antibiotics that cover enteric gram-negative organisms and anaerobes when appropriate. Early laparoscopic cholecystectomy is preferred in most surgical candidates. For patients who are too unstable or high-risk for immediate surgery, percutaneous cholecystostomy can decompress the gallbladder and serve as a bridge or definitive palliation.

5. Choledocholithiasis and Ascending Cholangitis

Choledocholithiasis

Choledocholithiasis means a stone is present in the common bile duct. A patient may be asymptomatic, may have jaundice and pruritus, or may present with biliary pancreatitis or cholangitis. The lab pattern is usually cholestatic: elevated direct bilirubin, alkaline phosphatase, and GGT. Transaminases can spike early if obstruction is abrupt, but persistent obstruction usually produces a ductal pattern. Ultrasound may show duct dilation but is less sensitive for visualizing common duct stones than for gallbladder stones.

The major clinical decision is whether the patient has low, intermediate, or high probability of a common duct stone. Low-risk patients can often proceed to cholecystectomy without preoperative ERCP. Intermediate-risk patients often need MRCP or endoscopic ultrasound to confirm duct stones. High-risk patients, especially those with cholangitis or a clearly obstructed duct, generally need therapeutic ERCP because ERCP can remove stones, perform sphincterotomy, and place stents.

Ascending cholangitis

Ascending cholangitis is infected biliary obstruction. Biliary pressure rises behind a blocked duct, bacteria proliferate in stagnant bile, and bacteremia or septic shock can develop rapidly. Charcot triad is fever, right upper quadrant pain, and jaundice. Reynolds pentad adds hypotension and altered mental status and signals severe disease. Absence of the complete triad does not exclude cholangitis, especially in older, immunocompromised, or critically ill patients.

Syndrome	Key diagnostic clue	Usual imaging/procedure	Management priority
Choledocholithiasis without infection	Jaundice/cholestatic labs; duct dilation; minimal fever	US → MRCP/EUS if intermediate risk; ERCP if high risk	Duct clearance; cholecystectomy to prevent recurrence
Acute cholangitis	Fever + jaundice/RUQ pain; sepsis physiology	Cultures and imaging; ERCP or PTC for drainage	Antibiotics + urgent biliary decompression
Gallstone pancreatitis	Lipase ≥3x ULN + gallstones or cholestatic labs	US for stones; ERCP only if ongoing obstruction/cholangitis	Supportive care; cholecystectomy during index admission when mild

Treatment of cholangitis begins with sepsis care: IV access, fluids, cultures, broad-spectrum antibiotics, hemodynamic monitoring, and early consultation for biliary drainage. Antibiotics alone may temporarily improve fever, but source control requires decompression when obstruction persists or severity is moderate-to-severe. ERCP is usually preferred when feasible; percutaneous transhepatic drainage is used when ERCP is not possible or anatomy prevents access.

6. Primary Sclerosing Cholangitis, Primary Biliary Cholangitis, and Other Duct Disorders

Primary sclerosing cholangitis

Primary sclerosing cholangitis (PSC) is a chronic cholestatic disease characterized by inflammation, fibrosis, and multifocal stricturing of intrahepatic and/or extrahepatic bile ducts. It has a strong association with ulcerative colitis and can progress despite colectomy. Patients may present with fatigue, pruritus, recurrent cholangitis, jaundice, or abnormal alkaline phosphatase discovered incidentally. MRCP or ERCP can show multifocal strictures and dilations that create a beaded duct appearance.

PSC is important because it links bowel inflammation to biliary malignancy. Complications include recurrent bacterial cholangitis, dominant strictures, secondary biliary cirrhosis, portal hypertension, cholangiocarcinoma, gallbladder carcinoma, and colorectal cancer risk in patients with associated IBD. Management is supportive and complication-focused; liver transplantation is the only curative treatment for advanced disease. Dominant strictures may require endoscopic dilation or stenting after malignancy is considered.

Primary biliary cholangitis

Primary biliary cholangitis (PBC) is an autoimmune cholestatic liver disease of small intrahepatic bile ducts, classically affecting middle-aged women. Symptoms include fatigue and pruritus; later disease can cause jaundice, xanthelasma, osteoporosis, fat-soluble vitamin deficiency, portal hypertension, and cirrhosis. The typical lab pattern is cholestatic, and antimitochondrial antibodies are positive in most patients. Ursodeoxycholic acid slows progression; obeticholic acid or fibrates may be considered in selected incomplete responders, and transplantation is used for end-stage disease.

Feature	PSC	PBC
Primary target	Large and medium intrahepatic/extrahepatic ducts	Small intrahepatic bile ducts
Typical association	Ulcerative colitis, recurrent cholangitis	Autoimmune disease, middle-aged women
Diagnostic clue	Beaded ducts on MRCP/ERCP; cholestatic labs	Positive antimitochondrial antibody; cholestatic labs
Cancer concern	Cholangiocarcinoma; gallbladder cancer; CRC if IBD	HCC risk in advanced cirrhosis
Treatment concept	No curative medication; treat strictures/pruritus/infection; transplant advanced disease	Ursodeoxycholic acid; treat pruritus/bone disease; transplant advanced disease

Other biliary disorders

Cholangiocarcinoma is an adenocarcinoma of the bile ducts. It often presents with painless obstructive jaundice, pruritus, weight loss, dark urine, pale stools, and cholestatic labs. A hilar tumor at the confluence of the right and left hepatic ducts is called a Klatskin tumor. PSC, choledochal cysts, liver flukes, chronic biliary inflammation, and some toxins increase risk. Resectability is the key determinant of outcome, but many cases are advanced at diagnosis.

Choledochal cysts are congenital cystic dilations of the biliary tree. They are more common in females and can cause abdominal pain, jaundice, fever, pancreatitis, cholangitis, biliary obstruction, cirrhosis, or cholangiocarcinoma. Because malignant transformation is a feared complication, definitive management usually involves surgical excision and biliary-enteric reconstruction.

Bile duct strictures can be iatrogenic after cholecystectomy or transplantation, inflammatory from chronic pancreatitis or PSC, malignant from pancreaticobiliary tumors, or recurrent from stones. A benign stricture may still produce life-threatening cholangitis because duct obstruction allows bacterial overgrowth. Endoscopic stenting is preferred when feasible; surgery is needed when obstruction is complete, recurrent, or not endoscopically manageable.

Biliary dyskinesia refers to biliary-type pain without gallstones or structural obstruction, often evaluated with HIDA scan after cholecystokinin stimulation. A low gallbladder ejection fraction supports impaired gallbladder emptying, but the diagnosis should be made cautiously because functional pain syndromes and sphincter of Oddi disorders can mimic biliary colic.

7. Gallbladder Cancer and Benign Gallbladder/Biliary Risk States

Gallbladder carcinoma is usually adenocarcinoma and often presents late with nonspecific right upper quadrant pain, weight loss, anorexia, jaundice, or a mass. Gallstones are common in affected patients, but gallstones are common in the population and most patients with stones never develop cancer. Porcelain gallbladder, chronic inflammation, large polyps, anomalous pancreaticobiliary junctions, and some infections increase risk. Prognosis is poor when disease is discovered after invasion or metastasis.

Porcelain gallbladder means calcification of the gallbladder wall. Historically it was considered a strong cancer risk marker, but modern estimates vary by calcification pattern. The practical teaching point remains that a calcified gallbladder should trigger careful surgical evaluation rather than being treated as a routine incidental stone finding.

8. Acute Pancreatitis

Definition and pathophysiology

Acute pancreatitis is acute pancreatic inflammation usually diagnosed when at least two of the following are present: characteristic epigastric pain, serum lipase or amylase at least three times the upper limit of normal, or imaging findings consistent with pancreatitis. Pain is classically severe, epigastric, persistent, and radiates to the back. Nausea, vomiting, ileus, fever, tachycardia, hypotension, hypoxemia, and third-spacing can occur. The most common etiologies in adults are gallstones and alcohol; additional causes include hypertriglyceridemia, hypercalcemia, medications, ERCP, trauma, infection, autoimmune disease, and malignancy.

Original diagram: pancreatitis mechanism and management logic

Trigger	Pancreatic response	Clinical consequence	Management concept
Gallstone, alcohol, triglycerides, drugs, ERCP	Premature enzyme activation and acinar injury	Inflammation, pain, third-spacing, SIRS	Analgesia, monitored fluids, early feeding
Necrosis or duct leak	Local collections form over days to weeks	Pseudocyst, infected necrosis, walled-off necrosis	Drain only when symptomatic, infected, or complicated

Severity depends more on organ failure than on the lipase value. Mild pancreatitis has no organ failure or local complications. Moderately severe pancreatitis has transient organ failure or local/systemic complications. Severe pancreatitis has persistent organ failure lasting more than 48 hours. Early warning signs include rising BUN, hemoconcentration, tachycardia, hypoxemia, hypotension, systemic inflammatory response, renal injury, and altered mental status.

Domain	High-yield anchor	Clinical use
Diagnosis	2 of 3: pain, lipase/amylase $\geq 3\times$ ULN, imaging	Avoid CT solely to prove classic pancreatitis early
Etiology workup	RUQ ultrasound for gallstones; triglycerides if no clear cause	Gallstone pancreatitis changes ERCP/cholecystectomy planning
Early fluids	Moderately aggressive crystalloid; lactated Ringer often preferred	Monitor vitals, urine output, BUN, hematocrit, cardiopulmonary risk
Nutrition	Early oral feeding with low-fat solid diet when tolerated in mild disease	Avoid prolonged bowel rest when symptoms improve
Antibiotics	Not routine for sterile pancreatitis	Use when infected necrosis or extrapancreatic infection is suspected
ERCP	Not routine in all gallstone pancreatitis	Use urgently for cholangitis or persistent biliary obstruction

Complications

Local complications include acute peripancreatic fluid collection, pancreatic pseudocyst, acute necrotic collection, walled-off necrosis, infected necrosis, splenic vein thrombosis, hemorrhage, pancreatic ascites, and gastric outlet or biliary obstruction. A pseudocyst is an encapsulated fluid collection rich in pancreatic enzymes that usually develops several weeks after pancreatitis. Many pseudocysts resolve spontaneously; drainage is considered for persistent symptoms, infection, bleeding, obstruction, enlargement, or complications.

Systemic complications include shock, acute kidney injury, hypoxemia/ARDS, pleural effusion, disseminated intravascular coagulation, hypocalcemia, hyperglycemia, and multiorgan failure. Grey Turner and Cullen signs are late signs of hemorrhagic pancreatitis and should not be awaited before escalating care.

9. Chronic Pancreatitis and Exocrine Pancreatic Insufficiency

Chronic pancreatitis is progressive fibroinflammatory destruction of the pancreas, most commonly associated with alcohol use in many populations but also linked to smoking, genetic variants, recurrent acute pancreatitis, duct obstruction, autoimmune pancreatitis, hypertriglyceridemia, and idiopathic disease. The core clinical pattern is chronic or recurrent epigastric pain radiating to the back, pancreatic calcifications, ductal irregularity, maldigestion, steatorrhea, weight loss, and pancreatogenic diabetes.

Exocrine pancreatic insufficiency occurs when enzyme output is inadequate for fat and protein digestion. Steatorrhea generally becomes obvious only after severe loss of exocrine reserve. Patients may have bulky, oily, foul-smelling stools, weight loss, fat-soluble vitamin deficiency, and low fecal elastase. Treatment includes alcohol and tobacco cessation, nutrition support, pancreatic enzyme replacement with meals and snacks, fat-soluble vitamin replacement, and pain management that avoids escalating opioid dependence when possible.

Condition	Dominant clue	Diagnostic tools	Treatment principles
Chronic pancreatitis	Chronic epigastric/back pain, calcifications, steatorrhea	CT/MRCP/EUS; fecal elastase for insufficiency	Stop alcohol/tobacco, enzymes, nutrition, pain plan, manage diabetes
Autoimmune pancreatitis	Painless jaundice or pancreatic enlargement, IgG4 disease	IgG4, imaging, biopsy when malignancy unclear	Steroids; relapse monitoring
Pancreatic pseudocyst	Persistent cystic collection after pancreatitis, usually >4 weeks	CT/MRCP/EUS	Observe if asymptomatic; drain if symptomatic/complicated
Pancreatic cancer	Painless jaundice, weight loss, new diabetes, Courvoisier sign	Pancreatic protocol CT, EUS biopsy	Resection if localized; biliary stent/palliation/systemic therapy if advanced

10. Pancreatic Cancer

Pancreatic ductal adenocarcinoma is often advanced at diagnosis because early symptoms are vague. Tumors in the pancreatic head can obstruct the common bile duct and present with painless jaundice, dark urine, pale stools, pruritus, weight loss, anorexia, and sometimes a palpable nontender gallbladder. Tumors in the body or tail may present later with pain radiating to the back, weight loss, new diabetes, or metastatic disease. Risk factors include smoking, chronic pancreatitis, long-standing diabetes, obesity, family history, hereditary syndromes, and some genetic mutations.

Evaluation usually begins with contrast-enhanced pancreatic protocol CT when suspicion is high. Endoscopic ultrasound with biopsy helps establish tissue diagnosis and assess local invasion. CA 19-9 can support monitoring but is not a screening test and can rise with benign obstruction. The key management distinction is resectable, borderline resectable, locally advanced unresectable, or metastatic disease. Surgical resection offers the only chance of cure, but only a minority of patients are candidates at presentation.

11. Acute Appendicitis

Pathophysiology and clinical pattern

Appendicitis begins with luminal obstruction, commonly from lymphoid hyperplasia or fecalith, followed by increased intraluminal pressure, bacterial overgrowth, venous congestion, ischemia, inflammation, and possible perforation. Early pain is visceral and periumbilical because the appendix is a midgut structure; later pain localizes to the right lower quadrant when the parietal peritoneum becomes inflamed. Anorexia is common. Nausea and vomiting often follow the onset of pain rather than preceding it. Fever and leukocytosis are supportive but not required.

Original diagram: appendicitis progression

Luminal obstruction	Rising pressure	Ischemia/inflammation	Perforation risk
Fecalith, lymphoid hyperplasia, tumor	Venous congestion and bacterial overgrowth	RLQ peritoneal irritation	Abscess, peritonitis, sepsis

Classic signs include McBurney point tenderness, rebound, guarding, Rovsing sign, psoas sign, and obturator sign. However, anatomy changes the presentation. A retrocecal appendix can cause flank or back pain, a pelvic appendix can cause urinary or rectal symptoms, pregnancy can shift pain location, and older adults may present with subtle symptoms despite advanced disease. Perforation risk rises with delayed treatment, extremes of age, immunosuppression, and atypical presentations.

Diagnosis and management

Appendicitis remains a clinical diagnosis supported by labs and imaging. CT abdomen/pelvis with IV contrast is highly accurate in adults with uncertain presentations. Ultrasound is often used in children and pregnant patients, although a nondiagnostic ultrasound does not exclude appendicitis. MRI is useful in pregnancy when ultrasound is nondiagnostic and expertise is available. Broad-spectrum antibiotics are given when appendicitis is suspected, especially if surgery is planned or perforation is possible.

Laparoscopic appendectomy is standard for many patients, especially complicated appendicitis, appendicolith, perforation, abscess not amenable to nonoperative management, diffuse peritonitis, or diagnostic uncertainty with high concern. Antibiotics-first management can be considered in carefully selected uncomplicated cases after shared decision-making, but recurrence and later surgery remain important counseling points. Abscess or phlegmon may require antibiotics, drainage, and interval surgical planning depending on course.

Appendicitis presentation	Clinical implication	Management direction
Uncomplicated, stable, no appendicolith/abscess/peritonitis	Nonoperative pathway may be reasonable in selected adults	Antibiotics-first or laparoscopic appendectomy after shared decision-making
Appendicolith present	Higher risk of failure/complication with antibiotics alone	Often favors appendectomy
Perforation, diffuse peritonitis, sepsis, free air	Complicated surgical abdomen	Resuscitation, antibiotics, urgent operative/source control
Abscess/phlegmon, stable	Localized complicated disease	Antibiotics +/- percutaneous drainage; interval planning

12. High-Yield Differentials and Mimics

Right upper quadrant pain has a broad differential. Gallbladder disease is common, but hepatitis, hepatic congestion, hepatic abscess, peptic ulcer disease, pancreatitis, right lower lobe pneumonia, pulmonary embolism, nephrolithiasis, pyelonephritis, herpes zoster, and myocardial ischemia can mimic biliary pain. Epigastric pain can represent peptic ulcer disease, pancreatitis, biliary colic, myocardial infarction, aortic disease, gastritis, esophagitis, or malignancy. Right lower quadrant pain can represent appendicitis, Crohn ileitis, ileocecal infection, renal colic, ovarian torsion, ectopic pregnancy, pelvic inflammatory disease, mesenteric adenitis, cecal diverticulitis, or testicular pathology.

Symptom/lab clue	Biliary disease clue	Important mimic to exclude
RUQ pain + fever + jaundice	Cholangitis	Sepsis from another source, hepatic abscess, viral hepatitis
Epigastric pain radiating to back	Pancreatitis	Aortic disease, MI, posterior ulcer perforation
Cholestatic labs without pain	Stone, stricture, PSC/PBC, cancer	Drug-induced liver injury, infiltrative disease
RLQ pain + anorexia	Appendicitis	Crohn disease, ovarian/testicular disease, stone
Painless jaundice + weight loss	Pancreaticobiliary malignancy	Benign stricture, PSC, choledocholithiasis

13. Initial Management Frameworks

Biliary sepsis pathway

- Recognize infected obstruction: fever, jaundice, RUQ pain, hypotension, altered mental status, leukocytosis, elevated bilirubin/ALP, duct dilation, or positive blood cultures.
- Resuscitate first: airway as needed, two IV lines, fluids, cultures, broad-spectrum antibiotics, lactate and organ function assessment.
- Do not stop at antibiotics when obstruction persists. Arrange source control with ERCP, percutaneous transhepatic drainage, or surgery depending on anatomy and stability.

Pancreatitis pathway

- Confirm diagnosis with symptoms, lipase/amylase $\geq 3\times$ upper limit of normal, or imaging. Do not overinterpret small enzyme elevations without a compatible syndrome.
- Assess etiology early: gallstones, alcohol, triglycerides, calcium, medications, ERCP, trauma, malignancy, autoimmune disease.
- Treat supportively: analgesia, monitored crystalloid resuscitation, antiemetics, oxygen if needed, early feeding when tolerated, and close reassessment for organ failure.

Appendicitis pathway

- Use clinical probability to decide imaging. In adults with uncertain presentation, CT with IV contrast is usually the most definitive initial imaging test.
- Give antibiotics when appendicitis is strongly suspected and involve surgery early for complicated disease or high-risk features.
- Consider antibiotics-first only for carefully selected uncomplicated cases after explaining recurrence and failure risks.

14. Common Pitfalls

- Calling cholangitis just gallstones. Gallstones are common; fever, jaundice, hypotension, mental status change, or organ dysfunction indicates infected obstruction and source control urgency.
- Using ERCP diagnostically in low-risk choledocholithiasis. ERCP is therapeutic and carries pancreatitis risk; MRCP or EUS is safer for intermediate probability.
- Giving prophylactic antibiotics for uncomplicated sterile pancreatitis. Antibiotics are for infected necrosis or extrapancreatic infection, not for enzyme elevation alone.
- Assuming lipase level predicts severity. Organ failure, hemodynamics, BUN/hematocrit trends, oxygenation, renal function, and necrosis are more important.
- Missing acalculous cholecystitis in critically ill patients because they cannot report typical RUQ pain.
- Treating jaundice without fractionating bilirubin and assessing cholestatic versus hepatocellular patterns.
- Over-relying on classic appendicitis signs. Retrocecal, pelvic, pregnant, elderly, diabetic, and immunosuppressed presentations are often atypical.

15. Practice Questions

A 44-year-old woman develops episodic post-prandial RUQ pain radiating to the right scapula. She has no fever, normal bilirubin, and gallstones on ultrasound. What process best explains the pain?

Answer: Biliary colic from transient cystic duct obstruction. The absence of fever, persistent pain, leukocytosis, and wall inflammation argues against acute cholecystitis.

A 71-year-old man presents with fever, jaundice, RUQ pain, hypotension, and confusion. Ultrasound shows a dilated common bile duct. What is the priority after initial stabilization?

Answer: Broad-spectrum antibiotics plus urgent biliary decompression, usually ERCP if feasible. This is severe ascending cholangitis, and source control is essential.

A patient with severe epigastric pain has a lipase four times the upper limit of normal. CT on arrival is normal. Does this exclude acute pancreatitis?

Answer: No. Acute pancreatitis can be diagnosed by characteristic pain plus lipase elevation. CT may be unnecessary early and can be falsely unrevealing before complications evolve.

A 32-year-old with ulcerative colitis has worsening pruritus and cholestatic labs. MRCP shows multifocal strictures and dilations of intrahepatic and extrahepatic ducts. What diagnosis links the bowel disease to the biliary findings?

Answer: Primary sclerosing cholangitis. PSC is strongly associated with UC and increases risk for cholangiocarcinoma and progressive biliary cirrhosis.

A 24-year-old has periumbilical pain that migrates to the RLQ with anorexia and pain on passive internal rotation of the flexed right hip. What mechanism explains the migration?

Answer: Early visceral midgut pain is referred to the periumbilical area; later inflammation of parietal peritoneum localizes pain to the RLQ.

16. Graph-RAG Extraction Map

Node	Class	High-value edges
Gallstone	Etiology / object	causes biliary colic; causes cholecystitis; causes choledocholithiasis; causes gallstone pancreatitis
Cystic duct obstruction	Mechanism	causes biliary colic if transient; causes acute cholecystitis if persistent
Common bile duct obstruction	Mechanism	causes cholestatic labs; causes obstructive jaundice; predisposes to cholangitis
Ascending cholangitis	Emergency syndrome	requires antibiotics; requires biliary decompression; presents with Charcot triad/Reynolds pentad
RUQ ultrasound	Diagnostic test	detects gallstones; detects duct dilation; supports cholecystitis
ERCP	Procedure / treatment	removes CBD stones; drains infected bile duct; can cause pancreatitis
Acute pancreatitis	Inflammatory disease	diagnosed by pain/lipase/imaging; caused by gallstones/alcohol; complicated by pseudocyst/necrosis/organ failure
PSC	Cholestatic liver disease	associated with UC; causes multifocal strictures; increases cholangiocarcinoma risk
PBC	Autoimmune cholestatic disease	associated with AMA; causes pruritus/cholestasis; treated with ursodiol
Appendicitis	Surgical abdomen	caused by luminal obstruction; progresses to perforation; diagnosed clinically plus imaging

17. References Used for Clinical Cross-Checking

American College of Gastroenterology. Management of Acute Pancreatitis guideline update, 2024. Key anchors used: diagnostic criteria, preference for monitored crystalloid resuscitation, no routine prophylactic antibiotics, early feeding in mild disease, and selective ERCP use.

Tokyo Guidelines 2018. Key anchors used: acute cholangitis/cholecystitis severity thinking, antibiotics plus biliary drainage/source control for acute cholangitis, and gallbladder drainage or surgery based on severity and operative risk.

American Society for Gastrointestinal Endoscopy guideline on choledocholithiasis risk stratification and endoscopic management. Key anchors used: reserve ERCP for high probability or therapeutic need; use MRCP/EUS for intermediate probability.

SAGES appendicitis guidance, 2024. Key anchors used: operative management remains favored for many patients, while antibiotics-first therapy may be considered in selected uncomplicated cases after shared decision-making.